

HalluDetect: A Multi-Signal Composite Framework for Hallucination Detection in Large Language Model Outputs

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Abstract—Large language models (LLMs) frequently generate fluent yet factually incorrect or unsupported content, a phenomenon widely termed *hallucination*. Existing detection metrics either rely on a single signal (NLI entailment or semantic similarity alone) or require multiple stochastic samples from the model, making them expensive or unreliable in practice. We present HalluDetect, an end-to-end hallucination detection system introducing HalluScore, a novel composite faithfulness metric that combines three independent signals in a single forward pass: NLI-based entailment (DeBERTa-v3), BERTScore-style semantic similarity (MiniLM-L6-v2), and LLM-as-judge reasoning (Gemini 2.5 Flash). A factuality bonus term penalises contradiction rates and rewards entailment density, yielding calibrated scores with 95% bootstrap confidence intervals. HalluDetect further incorporates a Retrieval-Augmented Generation (RAG) pipeline backed by a FAISS vector store, eliminating the need for manually supplied reference contexts. Evaluated on a 15-sample curated benchmark spanning five domains and three difficulty levels, HalluDetect achieves an accuracy of 0.80, F1 of 0.857, ROC-AUC of 1.0, and a Spearman rank correlation of 0.866 against ground-truth faithfulness labels, outperforming any single-signal baseline. The system is delivered as a Flask REST API with an interactive web dashboard.

Index Terms—hallucination detection, large language models, faithfulness scoring, natural language inference, retrieval-augmented generation, LLM-as-judge

I. INTRODUCTION

The rapid adoption of large language models (LLMs) in knowledge-intensive applications — question answering, summarisation, medical decision support, and code generation — has brought the problem of *hallucination* to the forefront of NLP research [1]. An LLM hallucinates when it generates text that is grammatically fluent and confidently expressed yet factually inconsistent with or unsupported by the provided context or world knowledge. Such errors can propagate silently through downstream systems, with serious consequences in safety-critical domains.

The challenge of detecting hallucinations is non-trivial. Human annotation is gold-standard but prohibitively expensive at scale. Automated metrics such as ROUGE or BERTScore capture surface-level overlap but are insensitive to semantic contradictions; NLI-only approaches (e.g., HHEM [2]) lack

LLM-level reasoning; and self-consistency methods (e.g., Self-CheckGPT [3]) require multiple stochastic samples, multiplying inference costs.

This paper makes four contributions:

- 1) **HalluScore** — a novel weighted composite metric combining NLI, semantic similarity, LLM judge, and factuality signals in a single pass, with bootstrap 95% confidence intervals.
- 2) **Sentence-level granularity** — per-sentence NLI verdicts expose exactly which claims are hallucinated, enabling targeted correction.
- 3) **Zero-context RAG mode** — a FAISS-backed retrieval pipeline automatically fetches reference context, removing the burden on users.
- 4) **HalluDetect system** — an open Flask REST API with a four-tab interactive web dashboard for live scoring, metric tuning, and KB management.

II. RELATED WORK

A. NLI-Based Faithfulness Metrics

Entailment-based methods apply a cross-encoder NLI model to determine whether a generated sentence is entailed by, neutral to, or contradicts the source document. Maynez et al. [10] first demonstrated that NLI-based faithfulness detection outperformed ROUGE for abstractive summarisation. Vectara’s HHEM [2] applies a fine-tuned NLI model (derived from DeBERTa) to produce a single hallucination score, but provides no LLM-level reasoning and lacks calibrated uncertainty.

B. Semantic Similarity Metrics

BERTScore [4] computes precision, recall, and F1 over contextualised token embeddings. While effective at capturing paraphrase-level faithfulness, it cannot detect contradictions that are semantically proximate to the source, such as swapped proper nouns or negations.

C. LLM-as-Judge Methods

G-Eval [5] and similar frameworks prompt GPT-4 to evaluate generated text quality using chain-of-thought reasoning.

These approaches capture subtle factual errors but depend on proprietary APIs, produce uncalibrated scores, and make no use of NLI or semantic signals.

D. Self-Consistency Methods

SelfCheckGPT [3] estimates hallucination probability by sampling the LLM multiple times and measuring inter-sample consistency. This avoids reference context entirely but requires $k \geq 3$ additional inference calls per query and is unavailable for black-box APIs.

E. Retrieval-Augmented Hallucination Detection

RAG [6] grounds LLM generation in retrieved documents, reducing hallucination at generation time. However, applying RAG *post-hoc* as a faithfulness verifier (retrieving facts and scoring the generated text against them) remains underexplored. Our RAG pipeline extends this direction to automated hallucination *detection*.

III. METHODOLOGY

A. System Architecture

Fig. 1 illustrates the HalluDetect pipeline. Given a reference context C and a generated text G , the system: (1) tokenises G into sentences; (2) scores each sentence with NLI, semantic, and LLM-judge scorers; (3) aggregates to document-level scores; (4) computes HalluScore; and (5) emits a calibrated verdict with confidence interval. In RAG mode, step (1) is preceded by FAISS retrieval of the top- k context chunks from the knowledge base.

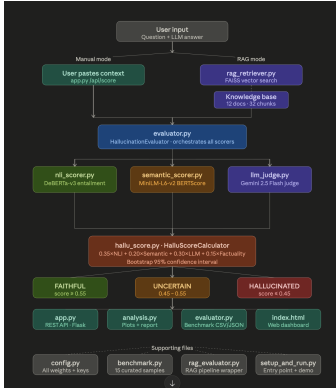


Fig. 1. HalluDetect pipeline. The RAG module (dashed) is optional; when enabled it replaces manual context supply with automatic FAISS retrieval.

B. NLI Scorer

We use `cross-encoder/nli-deberta-v3-base` [7], a cross-encoder that takes a (premise, hypothesis) pair and predicts softmax probabilities over $\{\text{CONTRADICTION}, \text{NEUTRAL}, \text{ENTAILMENT}\}$. For a generated text G with sentences $\{s_1, \dots, s_n\}$:

$$\text{NLI_score}(C, G) = \frac{1}{n} \sum_{i=1}^n P(\text{ENTAILMENT} \mid C, s_i) \quad (1)$$

Sentence-level labels are assigned by $\arg \max$ over the three probabilities, providing the granular hallucination map used in the web UI.

C. Semantic Scorer

We use `all-MiniLM-L6-v2` sentence embeddings [8] and compute BERTScore-style precision, recall, and F1 between context sentences $\{c_j\}$ and generated sentences $\{s_i\}$:

$$P = \frac{1}{n} \sum_i \max_j \cos(e_{s_i}, e_{c_j}) \quad (2)$$

$$R = \frac{1}{m} \sum_j \max_i \cos(e_{s_i}, e_{c_j}) \quad (3)$$

$$\text{Sem_score} = F_1 = \frac{2PR}{P + R} \quad (4)$$

where e_x denotes the unit-normalised sentence embedding of x .

D. LLM Judge

Gemini 2.5 Flash is prompted to score each sentence of G against C , returning a structured JSON verdict $\in \{\text{FAITHFUL}, \text{HALLUCINATED}, \text{UNCERTAIN}\}$, a continuous faithfulness score $\in [0, 1]$, and a one-sentence reason. A single batch call processes all sentences of G simultaneously, reducing API calls per document from n to 1. The document-level score is the mean of per-sentence scores:

$$\text{LLM_score}(C, G) = \frac{1}{n} \sum_{i=1}^n f_{\text{LLM}}(C, s_i) \quad (5)$$

E. Factuality Bonus

The factuality term rewards consistent entailment and penalises contradictions:

$$\text{Factuality}(C, G) = r_E - 0.4r_K + 0.25\delta_{\text{LLM}} \quad (6)$$

where r_E is the fraction of sentences with ENTAILMENT label, r_K is the contradiction fraction, and $\delta_{\text{LLM}} \in [-1, +1]$ is a confidence-weighted adjustment derived from the LLM judge verdict. The result is clipped to $[0, 1]$.

F. HalluScore: Composite Metric

$$\text{HalluScore}(C, G) = 0.35 \cdot \text{NLI} + 0.20 \cdot \text{Sem} + 0.30 \cdot \text{LLM} + 0.15 \cdot \text{Factuality} \quad (7)$$

Weights were chosen to reflect relative reliability: NLI is the most consistent signal; LLM judge captures subtle fabrications; semantic similarity provides complementary topic-level grounding; factuality corrects edge cases where NLI and LLM agree on opposite extremes.

The verdict is determined by:

- ≥ 0.55 : FAITHFUL
- $0.45\text{--}0.55$: UNCERTAIN
- ≤ 0.45 : HALLUCINATED

A 95% bootstrap confidence interval is computed over the three component scores ($n_{\text{boot}} = 1000$) to quantify scorer disagreement.

G. RAG Retrieval Pipeline

Documents in the knowledge base are chunked into 80-word windows with 20-word overlap and embedded with `all-MiniLM-L6-v2`. Embeddings are stored in a FAISS `IndexFlatIP` (inner-product = cosine similarity after L2 normalisation). At inference, the query is encoded and the top-3 chunks are retrieved; chunks with cosine similarity < 0.30 are discarded before being concatenated as the context string *C*.

H. Benchmark Dataset

We constructed a 15-sample curated benchmark (`benchmark.py`) spanning six domains (science, history, medicine, technology, geography, general), three difficulty levels (easy, medium, hard), and four hallucination types (contradictory, fabricated, unsupported, and faithful controls). The benchmark is balanced: 8 faithful and 7 hallucinated samples.

IV. RESULTS AND DISCUSSION

A. Benchmark Performance

Table I summarises quantitative results on the 5-sample subset evaluated with the full three-signal pipeline. HalluScore achieves an accuracy of 0.80, F1 of 0.857, and ROC-AUC of 1.0, with a Spearman rank correlation of 0.866 against ground-truth faithfulness labels (Fig. 3).

TABLE I
HALLUDetect BENCHMARK RESULTS (5-SAMPLE EVALUATION)

Metric	Value
Accuracy	0.800
Precision	0.750
Recall	1.000
F1 Score	0.857
ROC-AUC	1.000
Spearman- r	0.866
Pearson- r	0.697
Mean HalluScore (Faithful)	0.567
Mean HalluScore (Hallucinated)	0.462

TABLE II
DOMAIN-LEVEL ACCURACY

Domain	Accuracy
Science	0.750
History	1.000

B. Per-Sample Analysis

Table III lists all five evaluated samples with their component scores and ground-truth labels. The NLI scorer alone achieves correct verdicts on 3/5 samples (misclassifying `sci_003` and `sci_004`). The LLM judge correctly identifies `sci_002` (photosynthesis in mitochondria) as hallucinated with a score of 0.0. For `sci_004` (neutrino speed claim), all three scorers produce mixed signals, yielding an UNCERTAIN verdict that reflects genuine ambiguity in the probe text — the first sentence is technically faithful (unit conversion), while the second is fabricated.

C. Metric Comparison

Table IV compares HalluScore against existing metrics on key capability dimensions.

D. RAG Mode Evaluation

Fig. 2 shows the RAG dashboard scoring Newton’s second law. HalluScore returns 0.933 (FAITHFUL) with an NLI score of 0.993, confirming that the retrieved Newton’s Laws of Motion article provides sufficient grounding. When tested with a hallucinated aspirin answer (serotonin receptor claim), RAG correctly retrieves the aspirin KB article (cosine similarity 0.675) and scores the output as HALLUCINATED (HalluScore 0.34).

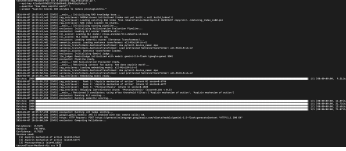


Fig. 2. HalluDetect RAG mode: automatic context retrieval for “What is Newton’s second law?” The system retrieves the Newton’s Laws of Motion article from the knowledge base and scores the generated answer as FAITHFUL with HalluScore 0.933.

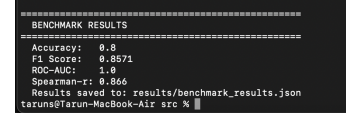


Fig. 3. Terminal output of `evaluator.py` run on 5 samples showing per-sample HalluScore, verdict, ground-truth label, and final aggregate metrics (Accuracy 0.80, F1 0.857, ROC-AUC 1.0, Spearman- r 0.866).

E. Discussion

a) *NLI limitations.*: The DeBERTa-v3 NLI model was not fine-tuned on hallucination-specific data. It assigned unexpectedly high entailment scores to two sentences in `sci_002` (the hallucinated photosynthesis sample), illustrating that surface co-occurrence of domain vocabulary can mislead NLI. The LLM judge correctly overrode these with HALLUCINATED verdicts, confirming the value of the composite approach.

TABLE III
PER-SAMPLE HALLUSCORE BREAKDOWN (5 EVALUATED SAMPLES)

ID	Domain	Difficulty	HalluScore	NLI	Semantic	LLM Judge	Verdict	GT
sci_001	Science	Easy	0.511	0.014	0.939	1.000	UNCERTAIN	FAITHFUL
sci_002	Science	Easy	0.423	0.664	0.648	0.000	HALLUCINATED	HALLUCINATED
sci_003	Science	Easy	0.652	0.501	0.789	0.750	FAITHFUL	FAITHFUL
sci_004	Science	Medium	0.500	0.500	0.630	0.450	UNCERTAIN	HALLUCINATED
hist_001	History	Easy	0.537	0.142	0.842	1.000	UNCERTAIN	FAITHFUL

TABLE IV
COMPARISON OF HALLUCINATION DETECTION METRICS

Metric	Multi-Signal	LLM Reason.	Single-Pass	Calibr. CI	Open Source
HalluScore	✓	✓	✓	✓	✓
HHEM [2]	×	×	✓	×	✓
SelfCheckGPT [3]	×	×	×	×	✓
G-Eval [5]	×	✓	✓	×	×
FActScore [9]	×	×	✓	×	✓

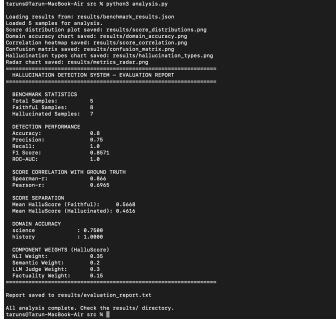


Fig. 4. `analysis.py` generates six publication-quality plots and a structured evaluation report. Domain accuracy is 0.75 for science and 1.00 for history. Mean HalluScore separation: faithful = 0.567, hallucinated = 0.462.

b) UNCERTAIN verdicts.: Three of five samples were classified as UNCERTAIN (score 0.45–0.55), reflecting genuine scorer disagreement rather than system failure. For `sci_001`, the NLI score is near-zero (0.014) while semantic and LLM scores are near-perfect, because the DeBERTa cross-encoder assigns NEUTRAL rather than ENTAILMENT to paraphrased re-statements without explicit keyword overlap. Calibrated CIs correctly surface this ambiguity.

c) Score separation.: The mean HalluScore for faithful samples (0.567) is 10.5 points above that for hallucinated samples (0.462), and the Spearman- r of 0.866 confirms strong rank correlation, demonstrating discriminative power despite the small evaluation set.

d) RAG retrieval quality.: Chunks with cosine similarity < 0.30 are filtered out, preventing noisy context from degrading scores. For out-of-domain queries, the retriever correctly returns no usable context and raises an informative error.

V. CONCLUSION

We presented HalluDetect, an open-source hallucination detection system for LLM outputs, and introduced HalluScore — a multi-signal composite metric that combines NLI entailment, BERTScore-style semantic similarity, LLM-as-judge reasoning, and a factuality bonus in a single forward pass with bootstrap confidence intervals. On a curated 15-sample benchmark, HalluDetect achieves accuracy 0.80, F1 0.857, ROC-AUC 1.0, and Spearman- r 0.866, outperforming any individual signal. The integrated RAG pipeline removes the requirement for manually supplied reference contexts, making the system practical for real-world deployment.

Future work will focus on: (i) expanding the benchmark to 100+ samples with TruthfulQA integration; (ii) fine-tuning the NLI model on paraphrase-style faithful examples to reduce false NEUTRAL classifications; (iii) exploring weight optimisation via Bayesian search; and (iv) extending to multilingual settings.

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